

Lab Guide 7

ConfigMap & Secret

Q.1 CREATE A YAML TO DEPLOY database with some env variable

```
MYSQL_ROOT_PASSWORD=redhat
```

```
MYSQL_DATABASE=mayank
```

```
MYSQL_USER=test1
```

```
MYSQL_PASSWORD =qwerty
```

Q.2 create a configmap with test1cm name and give 4 values as above

Q.3 create a secret with type generic and name of secret should be test1secret and give 4 values as above

Q.4 use test1cm configmap to create a pod using mysql image by yaml

Q.5 use test1secret secret to create a pod using mysql image by yaml

Ans: 1

```
kubectl run mydb --image mysql --env MYSQL_ROOT_PASSWORD=redhat --env  
MYSQL_DATABASE=mayank --env MYSQL_USER=test1 --env MYSQL_PASSWORD=qwerty --dry-run -o  
yaml > mydb.yaml
```

```
kubectl get pods
```

```
kubectl describe pod mydb
```

```
kubectl exec -it mydb -- bash
```

```
mysql -uroot -p enter
```

```
redhat
```

```
show databases;
```

```
env
```

```
exit
```

```

root@master:~# kubectl run mydb --image mysql --env MYSQL_ROOT_PASSWORD=redhat --env MYSQL_DATABASE=mayank --env MYSQL_USER=test1 --env MYSQL_PASSWORD=qwerty --dry-run -o yaml
W0524 05:27:18.221831 91930 helpers.go:663] --dry-run is deprecated and can be replaced with --dry-run=client.
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    run: mydb
    name: mydb
spec:
  containers:
  - env:
    - name: MYSQL_ROOT_PASSWORD
      value: redhat
    - name: MYSQL_DATABASE
      value: mayank
    - name: MYSQL_USER
      value: test1
    - name: MYSQL_PASSWORD
      value: qwerty
    image: mysql
    name: mydb
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
root@master:~#

```

```

root@master:~# kubectl get pods

```

NAME	READY	STATUS	RESTARTS	AGE
mydb	1/1	Running	0	18s
mysql	0/1	CrashLoopBackOff	5 (58s ago)	4m33s
mysql1	1/1	Running	0	3m27s

```

root@master:~#

```

```
root@master:~# kubectl describe pod mydb
Name: mydb
Namespace: jyothis
Priority: 0
Service Account: default
Node: node2/172.31.15.107
Start Time: Wed, 24 May 2023 05:28:32 +0000
Labels: run=mydb
Annotations: cni.projectcalico.org/containerID: ba10d2e453a9
              cni.projectcalico.org/podIP: 192.168.2.87/32
              cni.projectcalico.org/podIPs: 192.168.2.87/32
Status: Running
IP: 192.168.2.87
IPs:
  IP: 192.168.2.87
Containers:
  mydb:
    Container ID: containerd://72905def520a56d9818339ed8d0348ad
    Image: mysql
    Image ID: docker.io/library/mysql@sha256:a43f6e7e7f3a5e
    Port: <none>
    Host Port: <none>
    State: Running
      Started: Wed, 24 May 2023 05:28:46 +0000
    Ready: True
    Restart Count: 0
    Environment:
      MYSQL_ROOT_PASSWORD: redhat
      MYSQL_DATABASE: mayank
      MYSQL_USER: test1
      MYSQL_PASSWORD: qwerty
    Mounts:
```

```
root@master:~# kubectl exec -it mydb -- bash
bash-4.4#
bash-4.4#
bash-4.4# mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.33 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mayank |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> █
```

```

mysql> ^DBye
bash-4.4# env
MYSQL_PASSWORD=qwerty
HOSTNAME=mydb
MYSQL_DATABASE=mayank
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
MYSQL_ROOT_PASSWORD=redhat
KUBERNETES_PORT=tcp://10.96.0.1:443
PWD=/
HOME=/root
MYSQL_MAJOR=8.0
GOSU_VERSION=1.16
MYSQL_USER=test1
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT_443_TCP_PORT=443
MYSQL_VERSION=8.0.33-1.el8
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
TERM=xterm
SHLVL=1
KUBERNETES_SERVICE_PORT=443
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
KUBERNETES_SERVICE_HOST=10.96.0.1
MYSQL_SHELL_VERSION=8.0.33-1.el8
_=/usr/bin/env
bash-4.4# s

```

ANs: 2

vim mayankenv

cat mayankenv

kubectl create configmap test1cm --from-env-file mayankenv

kubectl get cm

kubectl describe cm test1cm

```

root@master:~# vim mayankenv
root@master:~# cat mayankenv
MYSQL_ROOT_PASSWORD=redhat
MYSQL_DATABASE=mayank
MYSQL_USER=test1
MYSQL_PASSWORD=qwerty
root@master:~#

```

```

root@master:~# kubectl create configmap test1cm --from-env-file mayankenv
configmap/test1cm created
root@master:~# kubectl get cm
NAME          DATA   AGE
kube-root-ca.crt  1       54m
test1cm        4       2s
root@master:~# kubectl describe cm test1cm
Name:         test1cm
Namespace:    jyothis
Labels:       <none>
Annotations:  <none>

Data
====
MYSQL_DATABASE:
-----
mayank
MYSQL_PASSWORD:
-----
qwerty
MYSQL_ROOT_PASSWORD:
-----
redhat
MYSQL_USER:
-----
test1

```

Ans: 3

```
kubectl create secret generic test1secret --from-env-file mayankenv
```

```
kubectl get secret
```

```
kubectl describe secret test1secret
```

```
kubectl edit secret test1secret
```

```

root@master:~# kubectl create secret generic test1secret --from-env-file mayankenv
secret/test1secret created
root@master:~# kubectl get secret
NAME          TYPE      DATA   AGE
test1secret    Opaque    4       3s
root@master:~# kubectl describe secret test1secret
Name:         test1secret
Namespace:    jyothis
Labels:       <none>
Annotations:  <none>

Type: Opaque

Data
====
MYSQL_DATABASE:      6 bytes
MYSQL_PASSWORD:      6 bytes
MYSQL_ROOT_PASSWORD: 6 bytes
MYSQL_USER:          5 bytes
root@master:~# kubectl edit secret test1secret
Edit cancelled, no changes made.
root@master:~# echo cmVkaGF0 | base64 -d
redhatroot@master:~# █

```

```
#
apiVersion: v1
data:
  MYSQL_DATABASE: bWF5YW5r
  MYSQL_PASSWORD: cXd1cnR5
  MYSQL_ROOT_PASSWORD: cmVkaGF0
  MYSQL_USER: dGVzdDE=
kind: Secret
metadata:
  creationTimestamp: "2023-05-24T05:35:57Z"
  name: test1secret
  namespace: jyothis
  resourceVersion: "22880"
  uid: df9a182d-59ed-4fb5-87fb-eb6f7ef62131
type: Opaque
~
~
```

Ans: 4

vim cmmydb.yaml

cat cmmydb.yaml

kubectl create -f cmmydb.yaml

kubectl get pods

kubectl describe pod cmmydb

kubectl exec -it cmmydb -- env

kubectl exec -it cmmydb -- mysql -uroot -p

```

root@master:~# cp mydb.yaml
root@master:~# cp mydb.yaml cmmydb.yaml
root@master:~# vim cmmydb.yaml
root@master:~# kubectl create -f cmmydb.yaml
pod/cmmydb created
root@master:~# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
cmmydb        1/1     Running   0           3s
mydb          1/1     Running   0           11m
mysql         0/1     CrashLoopBackOff   7 (4m32s ago)    16m
mysql1        1/1     Running   0           15m
root@master:~# kubectl describe pod cmmydb
Name:          cmmydb
Namespace:     jyothis
Priority:       0
Service Account: default
Node:          node2/192.31.15.107
Start Time:    Wed, 24 May 2023 05:40:24 +0000
Labels:        run=mydb
Annotations:   cnl.projectcalico.org/containerID: 812f789f015836dfb14b2370d26095fc5371b8f91080bb
               cnl.projectcalico.org/podIP: 192.168.2.88/32
               cnl.projectcalico.org/podIPs: 192.168.2.88/32
Status:        Running

```

```

root@master:~# cat cmmydb.yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    run: mydb
  name: cmmydb
spec:
  containers:
  - envFrom:
    - configMapRef:
        name: test1cm
    image: mysql
    name: mydb
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
root@master:~#

```

```

root@master:~# kubectl exec -it cmmydb -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=cmmydb
GOSU_VERSION=1.16
MYSQL_MAJOR=8.0
MYSQL_VERSION=8.0.33-1.el8
MYSQL_SHELL_VERSION=8.0.33-1.el8
MYSQL_PASSWORD=qwerty
MYSQL_ROOT_PASSWORD=redhat
MYSQL_USER=test1
MYSQL_DATABASE=mayank
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
TERM=xterm
HOME=/root
root@master:~# kubectl exec -it cmmydb -- mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.33 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> show databases;

```

Ans: 5

vim secretmydb.yaml

cat secretmydb.yaml

kubectl create -f secretmydb.yaml

kubectl get pods

kubectl describe pod secretmydb

kubectl exec -it secretmydb -- env

kubectl exec -it secretmydb -- mysql -uroot -p

```

root@master:~# vim secretmydb.yaml
root@master:~# kubectl create -f secretmydb.yaml
pod/secretmydb created
root@master:~# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
cmmydb        1/1     Running   0           3m1s
mydb          1/1     Running   0           14m
mysql         0/1     CrashLoopBackOff   8 (2m15s ago)   19m
mysql1        1/1     Running   0           18m
secretmydb    1/1     Running   0           5s
root@master:~# kubectl describe pod secretmydb
Name:          secretmydb
Namespace:     jyothis
Priority:       0
ServiceAccount: default
Node:          node3/172.31.9.49
Start Time:    Wed, 24 May 2023 05:43:20 +0000
Labels:        run=mydb
Annotations:    cnf.projectcalico.org/containerID: f0d6967424329f0cb7c294d269b94ac9cd4087137635b6bcb0817122a99ddaf5
                cnf.projectcalico.org/podIP: 192.168.3.82/32
                cnf.projectcalico.org/podIPs: 192.168.3.82/32
Status:        Running
IP:            192.168.3.82

```

```

root@master:~# kubectl exec -it secretmydb -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=secretmydb
GOSU_VERSION=1.16
MYSQL_MAJOR=8.0
MYSQL_VERSION=8.0.33-1.el8
MYSQL_SHELL_VERSION=8.0.33-1.el8
MYSQL_USER=test1
MYSQL_DATABASE=mayank
MYSQL_PASSWORD=qwerty
MYSQL_ROOT_PASSWORD=redhat
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
TERM=xterm
HOME=/root
root@master:~# kubectl exec -it secretmydb -- mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.33 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```